

Genomic Medicine & Course wrap-up

BFX-Workshop

Applied Bioinformatics for Genomics

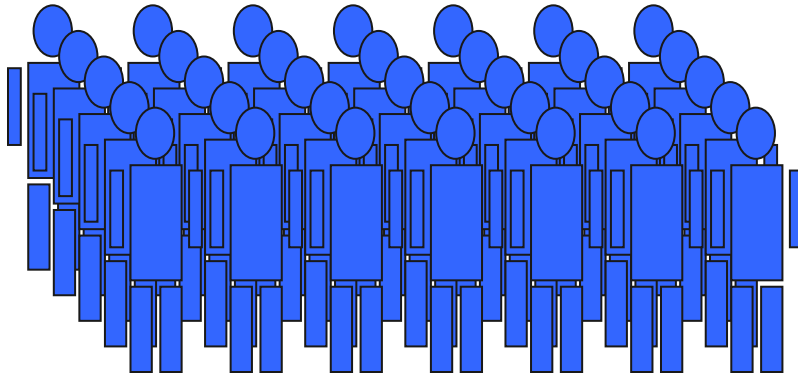
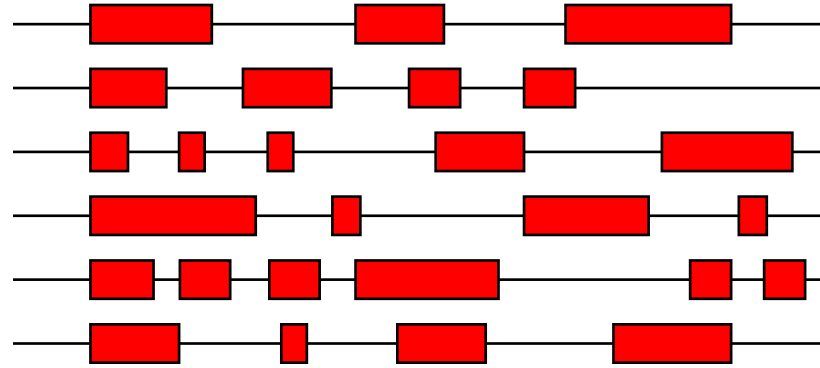
Chris Miller, Ph.D.

Pre-genomics cancer treatment

- Try Drug A
- If they don't respond, try Drug B
- If they don't respond, try Drug C
- ...

Targeted sequencing

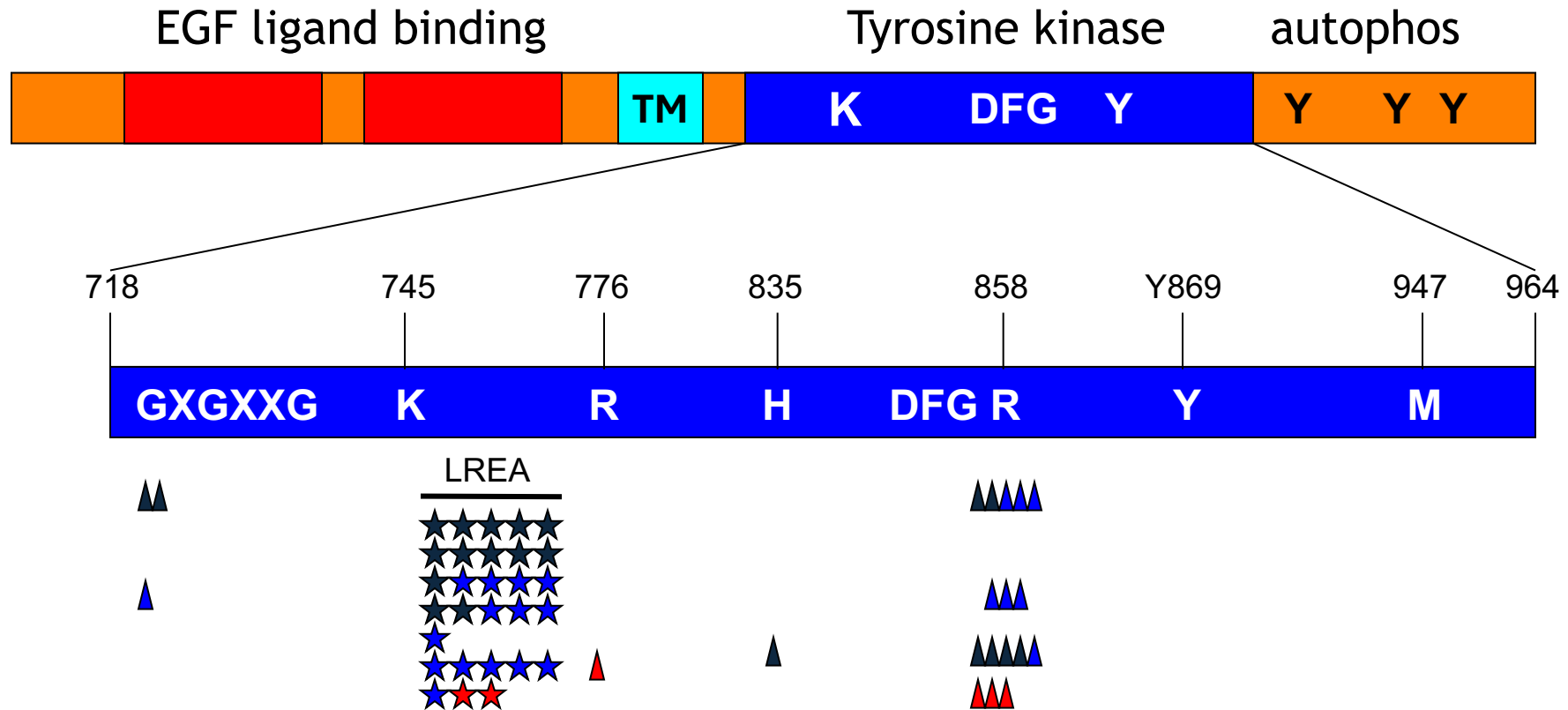
list of
candidate
genes



large collection of
patient samples

responders and
non-responders

EGFR mutations in never-smoker lung cancer



~80% of Iressa responders have EGFR mutations



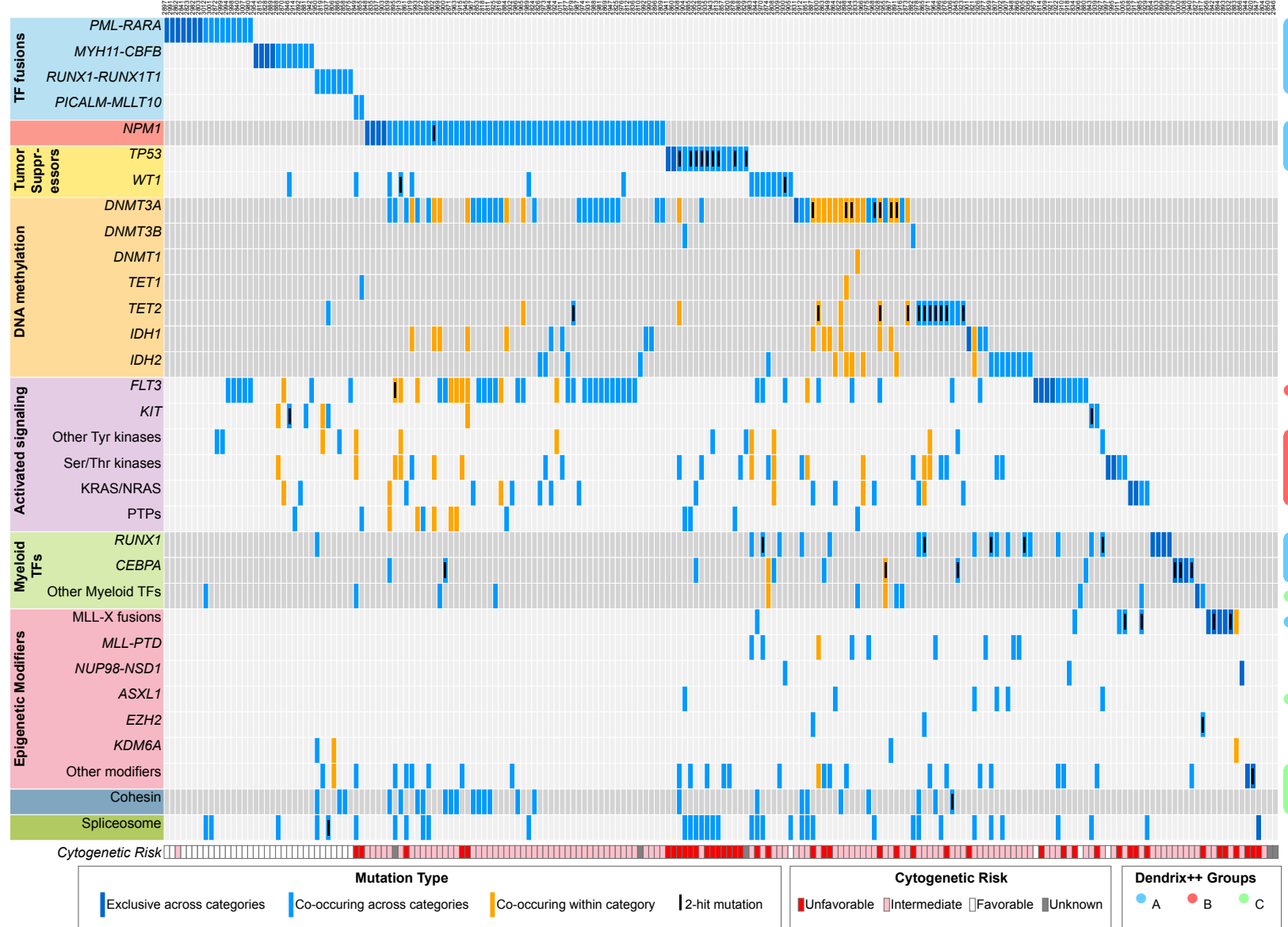
Personalized Genomic Medicine

- Using genomic tests to choose the right treatment for the right patient at the right time

Personalized Genomic Medicine

- Using genomic tests to choose the right treatment for the right patient at the right time
- Cancer is not one disease!

Recurrent somatic mutations in 200 AML patients



Lots of other poorly-defined diseases

- Multiple originating mutations giving rise to similar phenotypes
 - Cancer
 - Autosomal recessive deafness
 - Irritable bowel syndrome
 - many neurodevelopmental disorders

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- Multiple originating mutations giving rise to similar phenotypes
 - Cancer
 - Autosomal recessive deafness
 - Irritable bowel syndrome
 - many neurodevelopmental disorders
- Complex diseases with both environmental and genetic causes
 - Heart disease, diabetes, etc
 - Doses/metabolism

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Lots of other poorly-defined diseases

- Multiple originating mutations giving rise to similar phenotypes
 - Cancer
 - Autosomal recessive deafness
 - Irritable bowel syndrome
 - many neurodevelopmental disorders
- Complex diseases with both environmental and genetic causes
 - Heart disease, diabetes, etc
 - Doses/metabolism
- Treatments often work for only a small portion of the disease population

Genomic Medicine?

FDA Approves First Gene Therapies to Treat Patients with Sickle Cell Disease

Trikafta safe, effective for treating CF in analysis of clinical trials

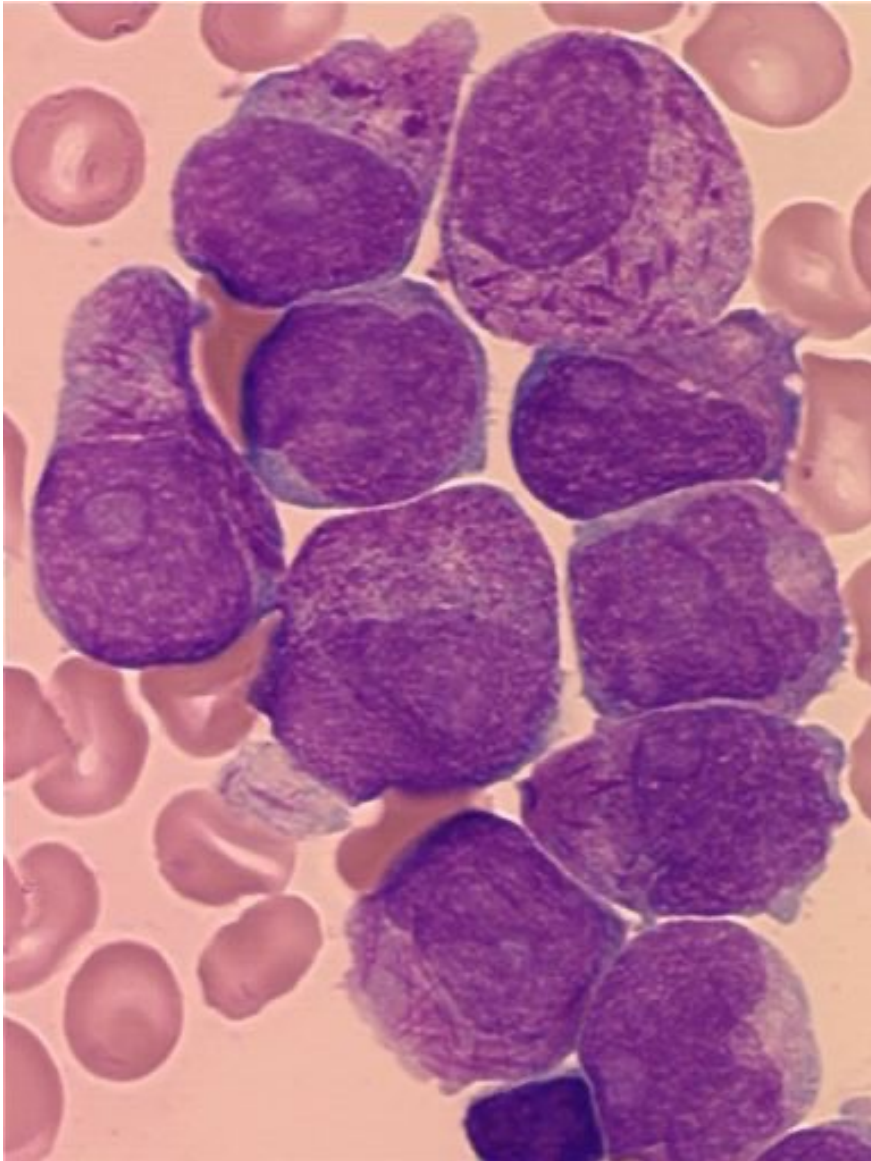
Researchers compared therapy against 2 other CFTR modulators, placebo

Certainly genomics played a role!

Gene Therapy Shows Promise in Initial Trial for Patients with Childhood Blindness

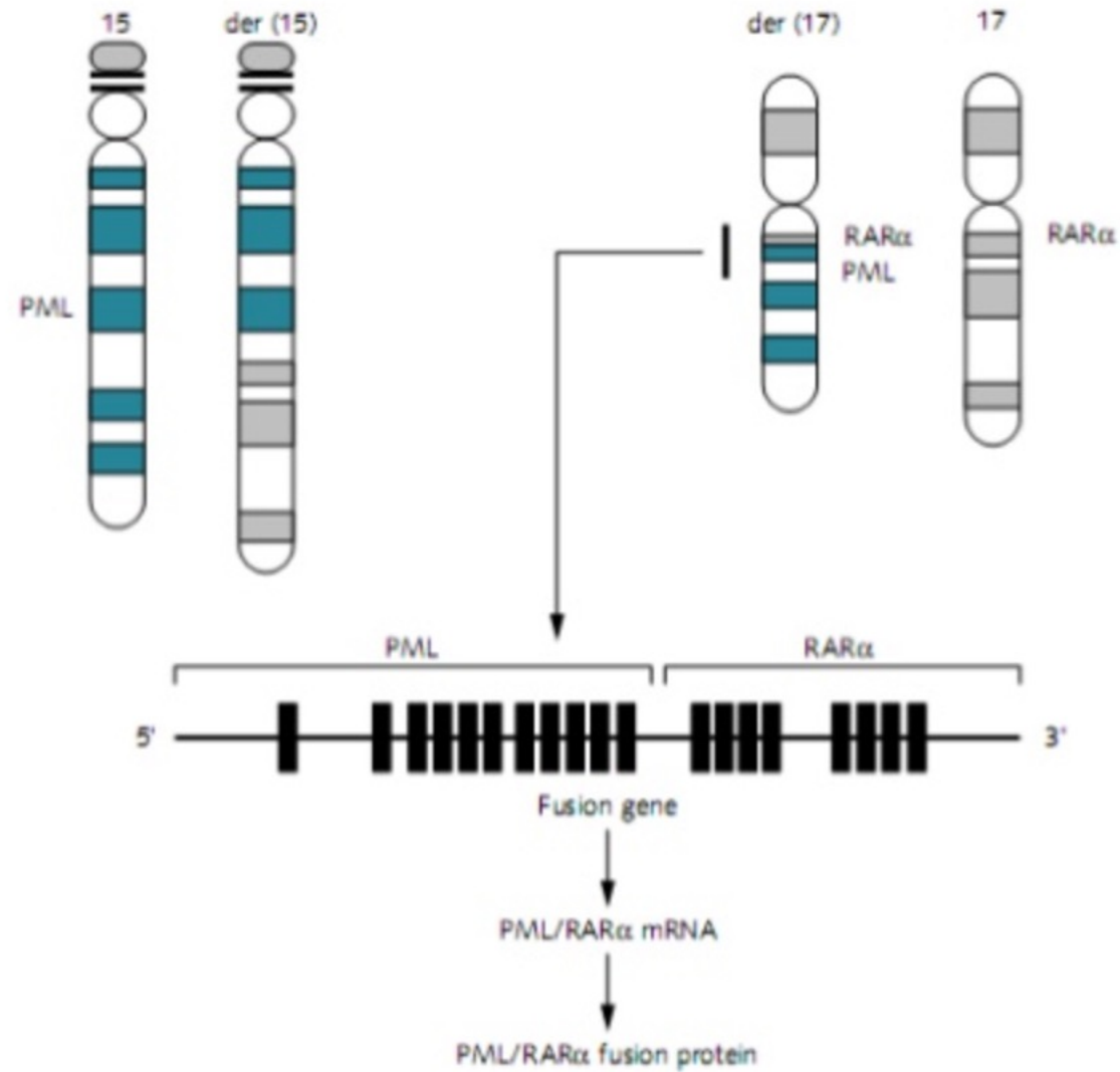
Penn Medicine researchers delivered working copies of the gene GUCY2D to the eyes of patients with severe vision impairments

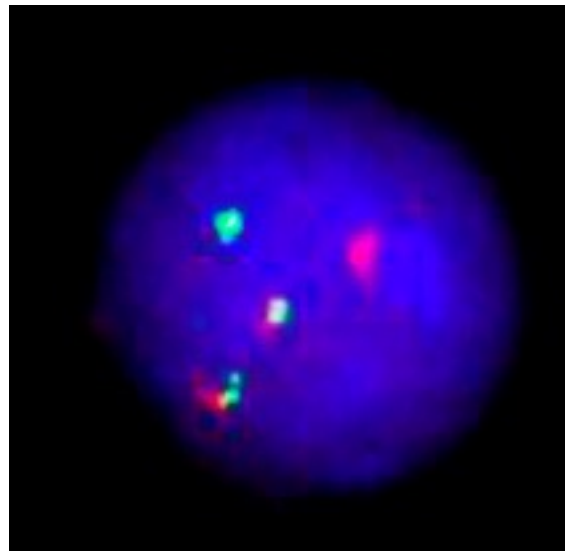
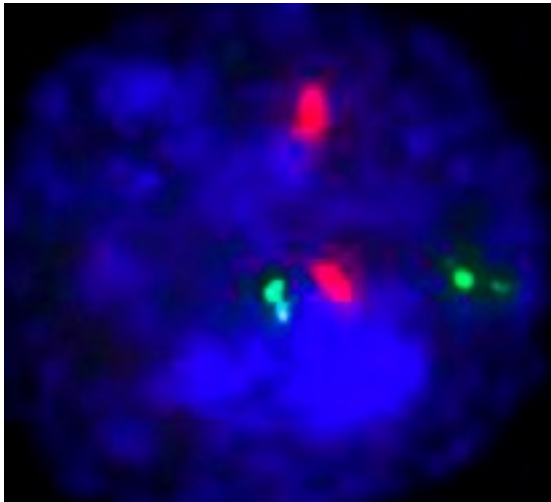
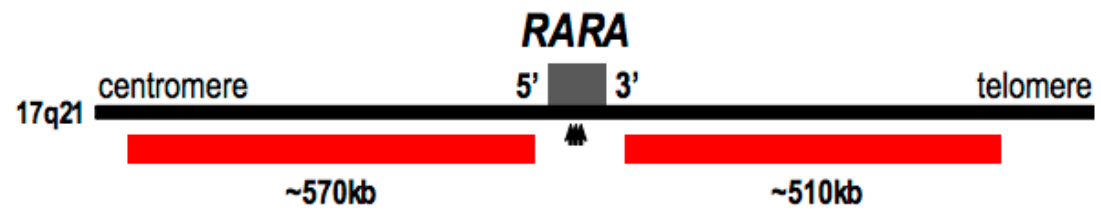
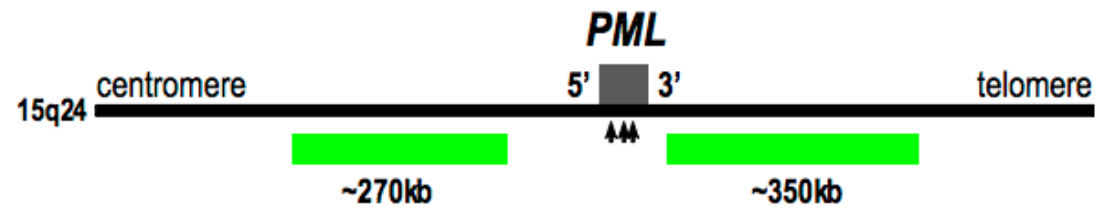
AML52: An atypical M3 AML



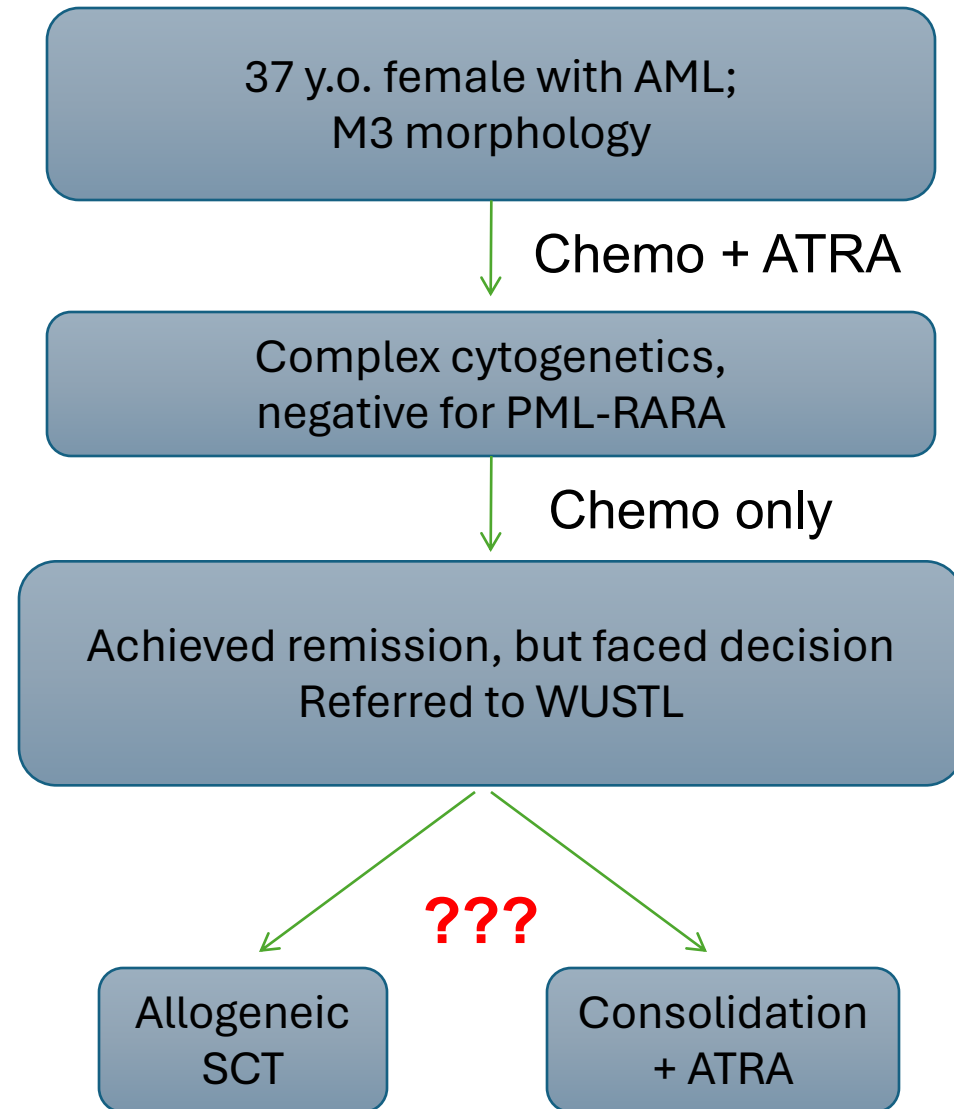
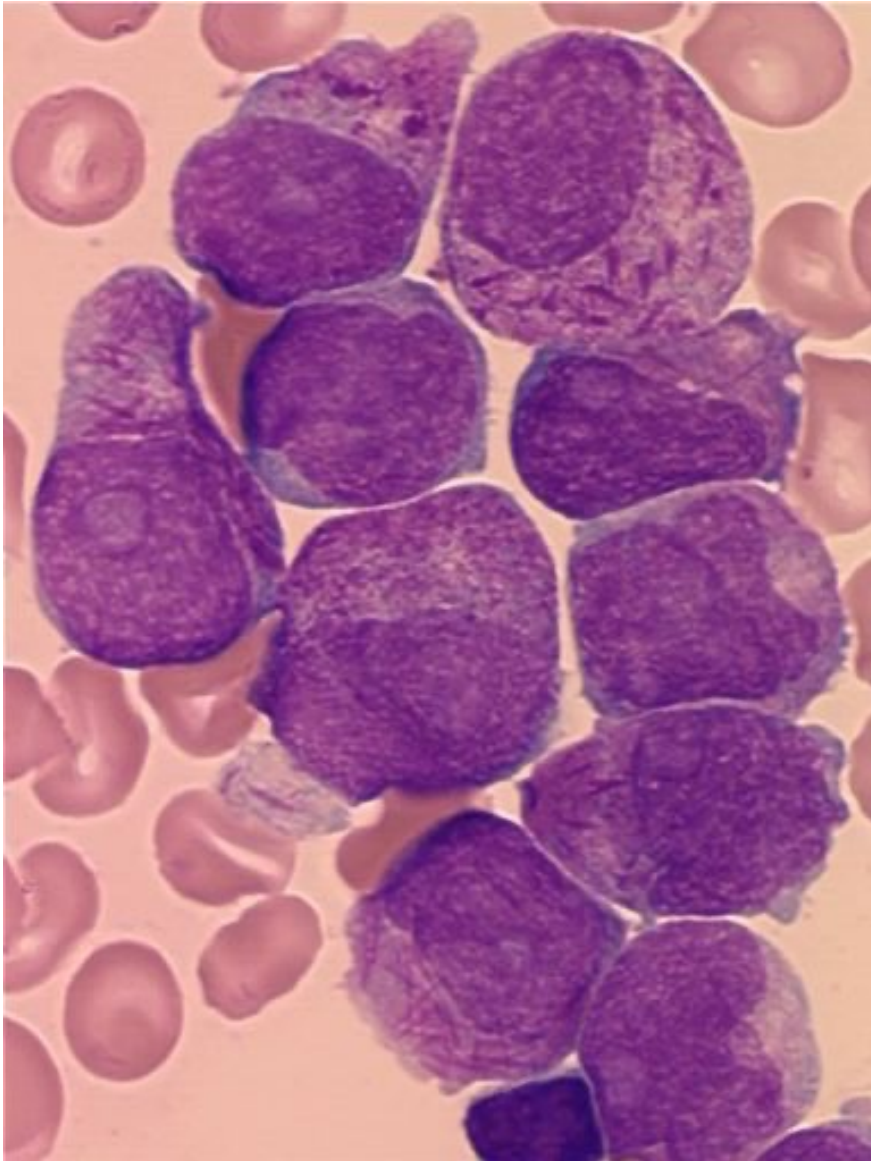
37 y.o. female with AML;
M3 morphology

PML-RARA fusion



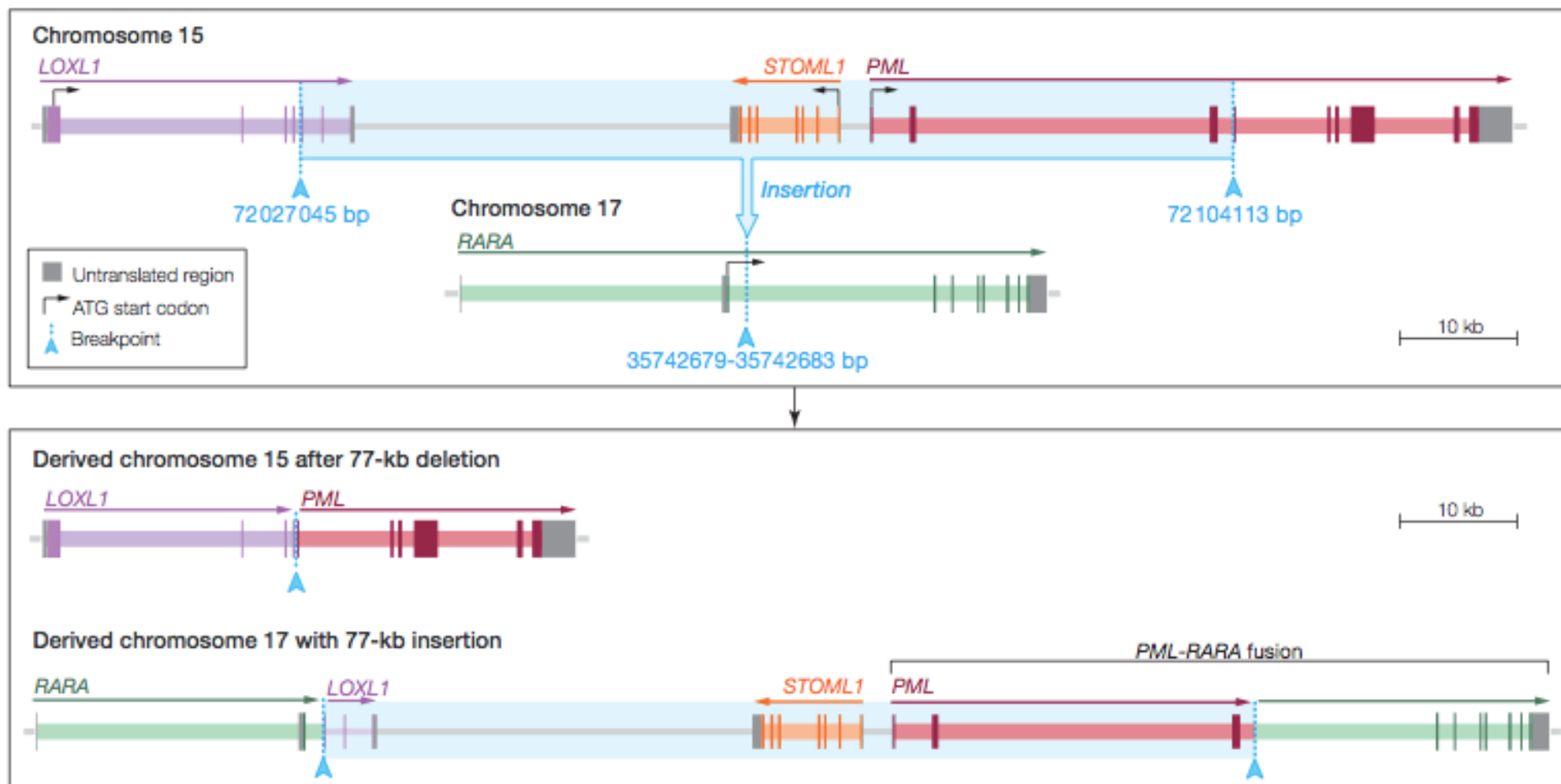


AML52: An atypical M3 AML

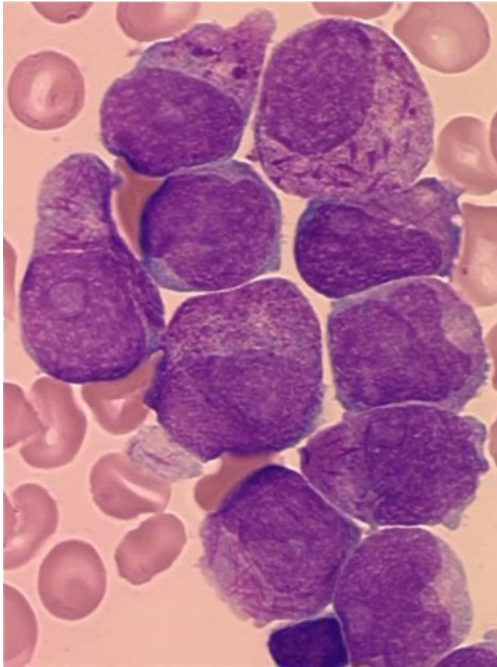


Use of Whole-Genome Sequencing to Diagnose a Cryptic Fusion Oncogene

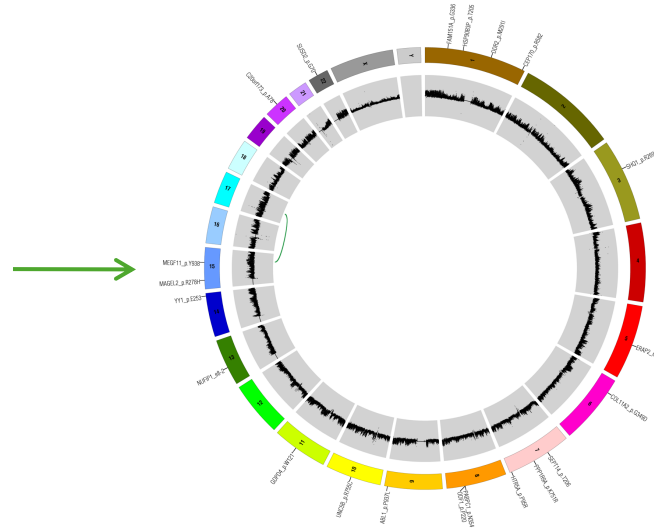
A Breakpoints in chromosomes 15 and 17 resulting in *PML-RARA* fusion



AML52: An atypical M3 AML



37 y.o. female with
de novo AML,
M3 morphology,
CTG, no PML-
RARA.
Referred to WUSM
for SCT.

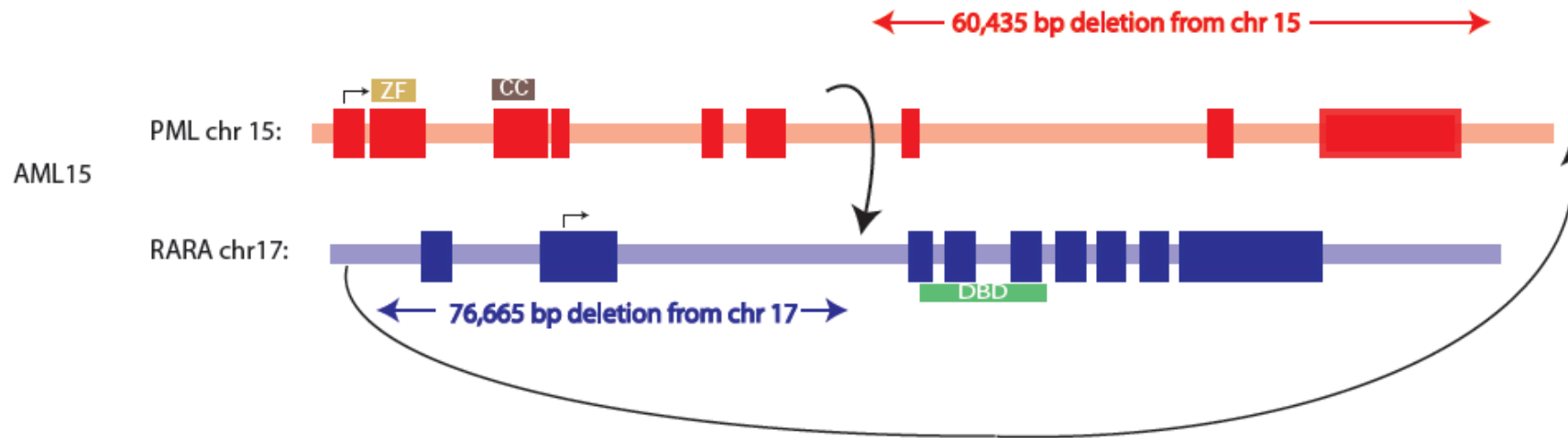
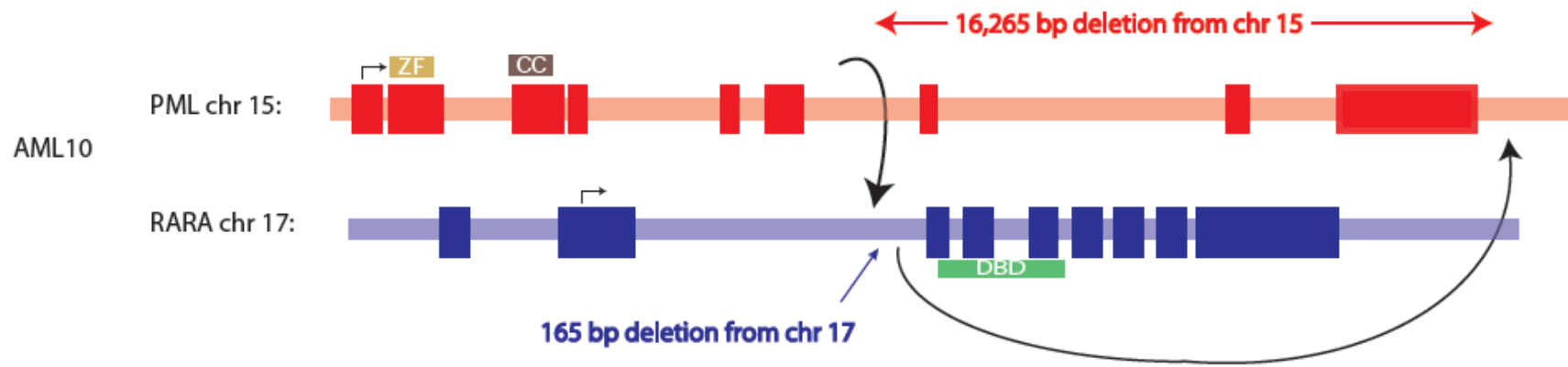


Detection of
PML-RARA
fusion by WGS.
Confirmed by
FISH, RT-PCR

Consolidation:
Chemo + ATRA

Sustained remission

Additional cryptic M3 AMLs



The New York Times

In Treatment for Leukemia, Glimpses of the Future



Second Chance: Lukas Wartman, a leukemia doctor and researcher, developed the disease himself. As he faced death, his colleagues sequenced his cancer genome. The result was a totally unexpected treatment.

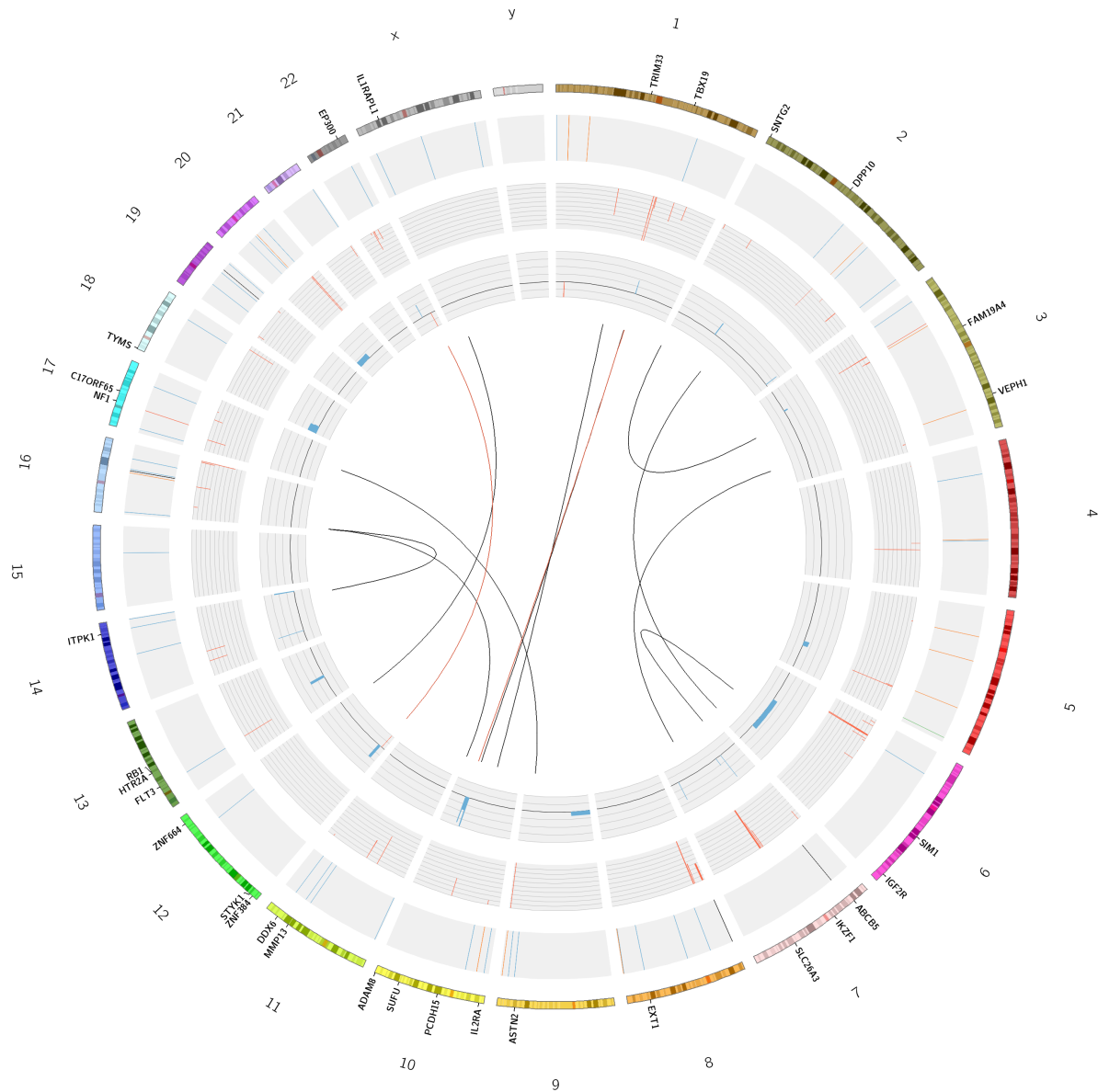
By **GINA KOLATA**

Published: July 7, 2012

A case study in personalized medicine

- Diagnosed with acute lymphocytic leukemia (B-ALL) - age 25
 - chemotherapy, achieved remission
- Relapse - age 30,
 - salvage chemo to remission; Stem cell transplant
- Severe 2nd relapse in July 2011 - age 33
 - CNS involvement.
- Whole genome, exome and RNA sequencing
 - initiated on August 1, 2011,
 - completed by early Sept 2011.

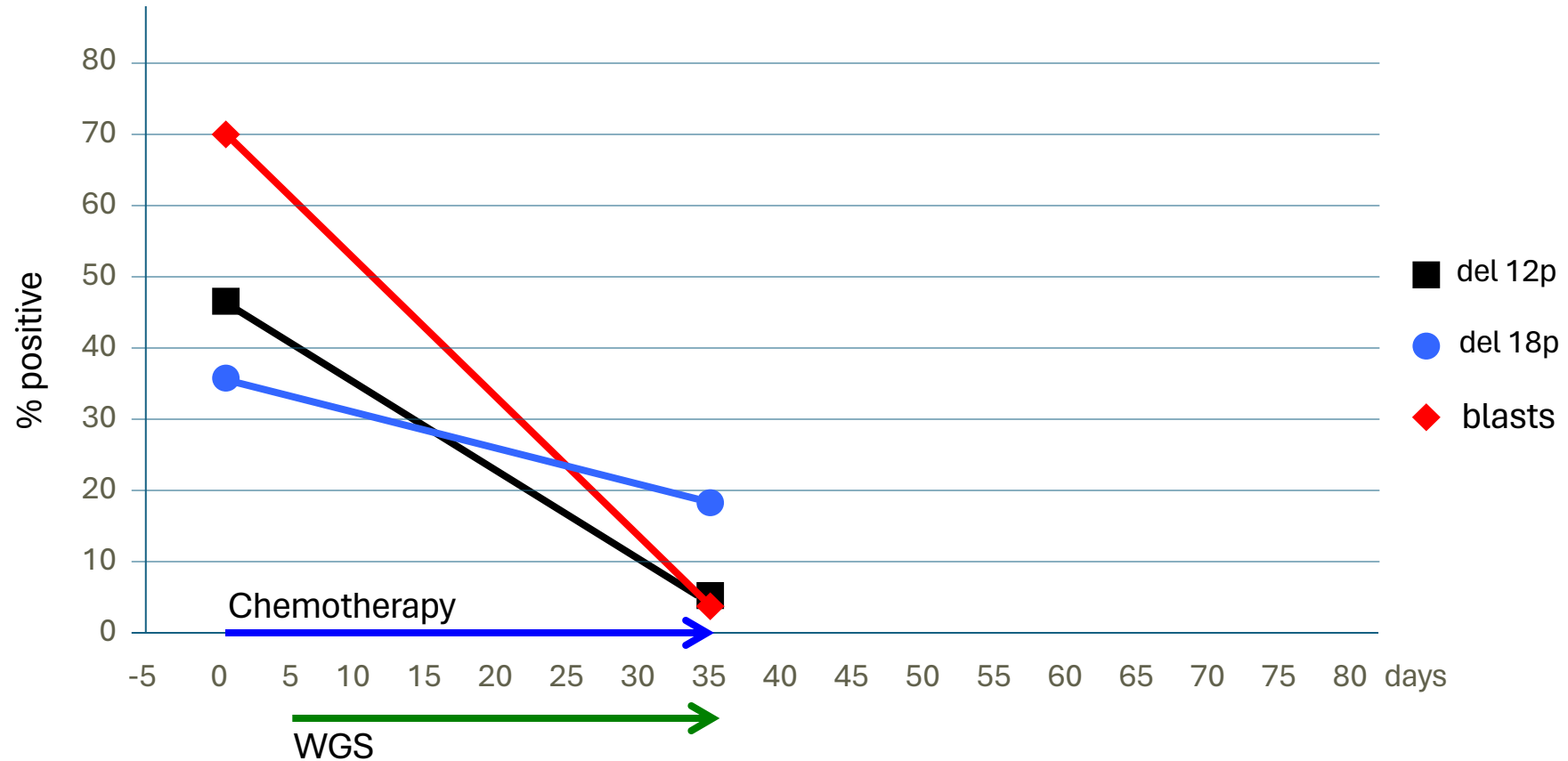
ALL1 genome summary (WGS/Exome/RNA-Seq)



Notable mutations

- *EP300*
- *NF1*
- *EXT1*
- *DDX6*
- *IL2RA*
- *MMP13*

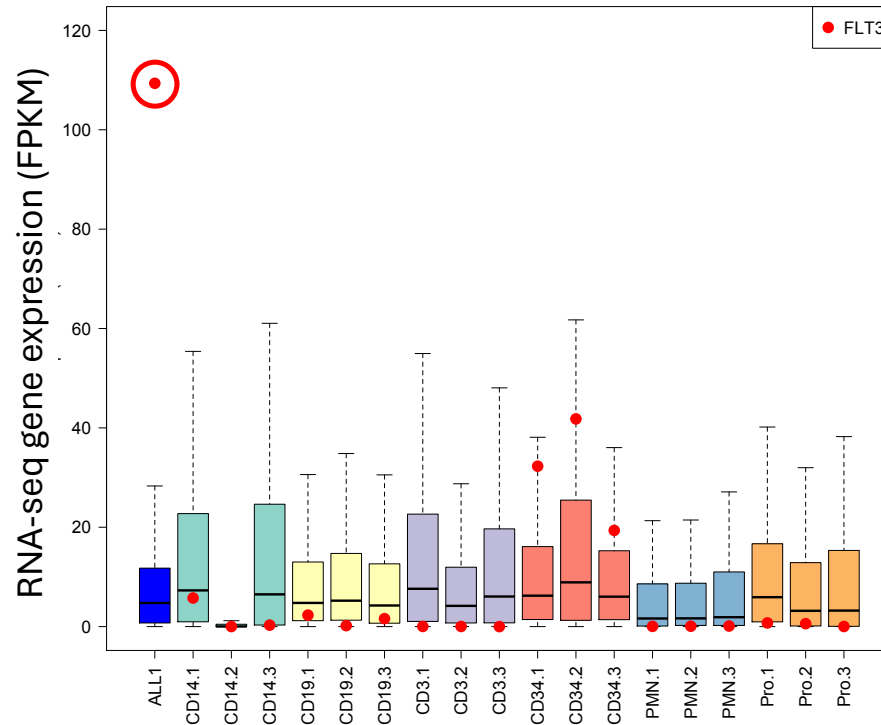
FISH at apparent remission



- By conventional pathology, the patient appeared to be in remission following salvage chemotherapy.
- The more highly sensitive i-FISH analysis indicated presence of treatment refractory disease. Hence, the patient was unable to receive transplant.

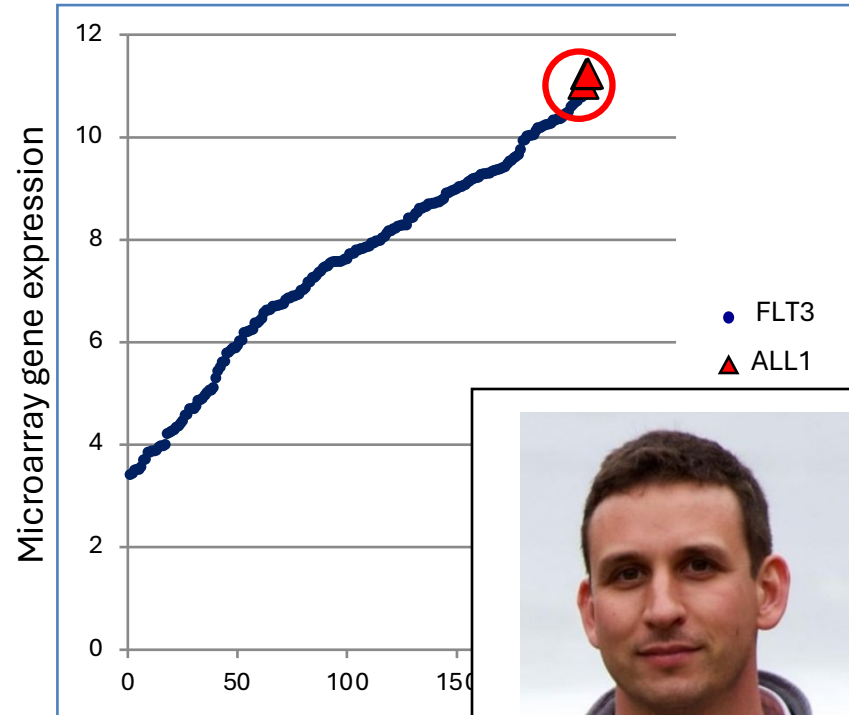
Extreme *FLT3* over-expression

RNA-seq expression data



ALL1 and 18 healthy donor samples

Microarray expression data



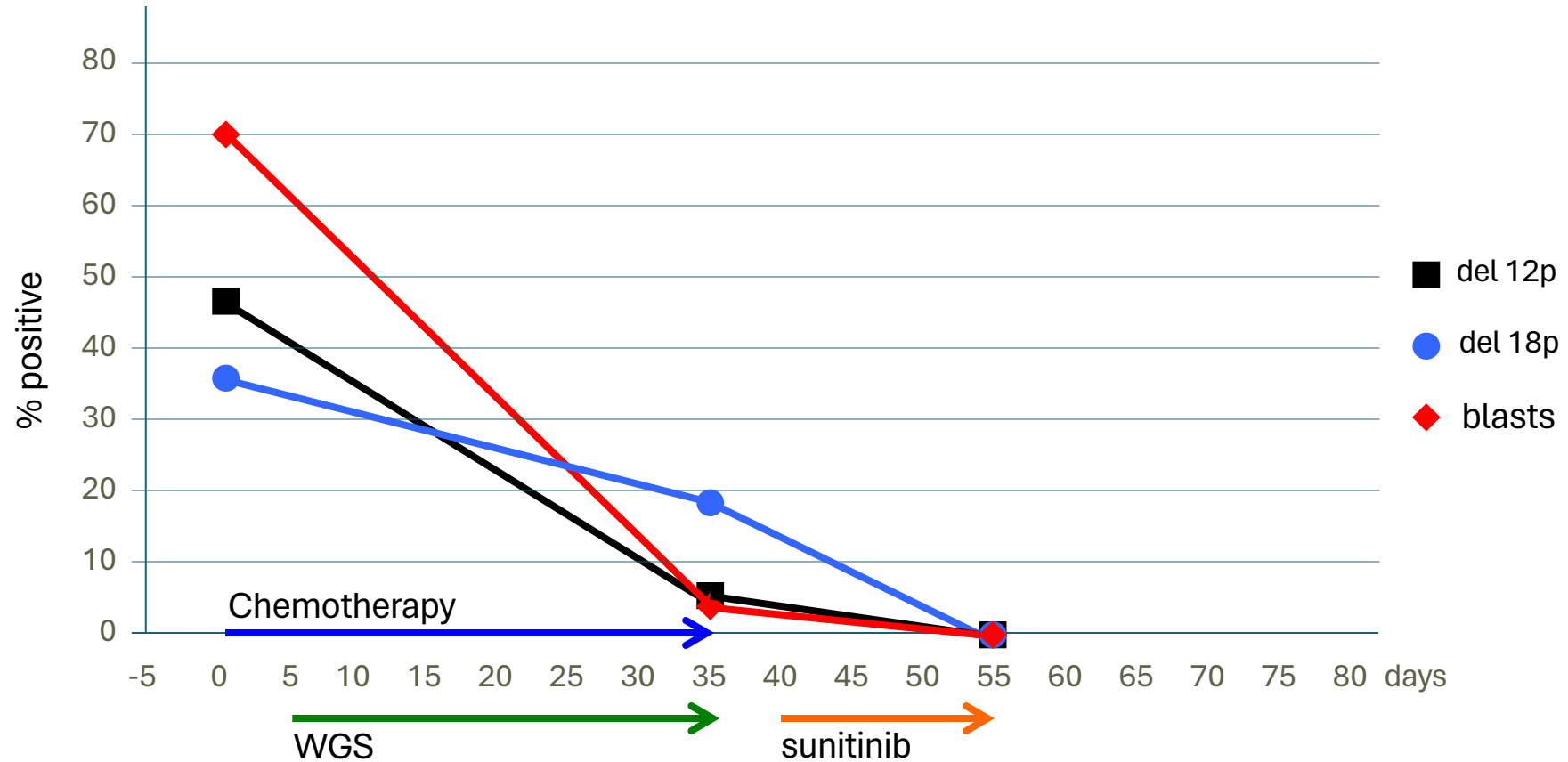
ALL1 (triplicates) and 18 healthy donor samples



Malachi Griffith

★ Targeted therapeutic for activated FLT3: sunitinib

FISH after targeted therapy



- No tumor cells detected following a 12-day course of sunitinib.
- Patient was able to receive a transplant to complete his treatment.

A case study in personalized medicine

- Sunitinib created a complete clinical remission in 12 days, enabling SCT (September 28, 2011).
- Genomic markers provided therapeutic options and better detection of residual disease
- Today: more than thirteen years post-SCT, Dr. Wartman is still alive!

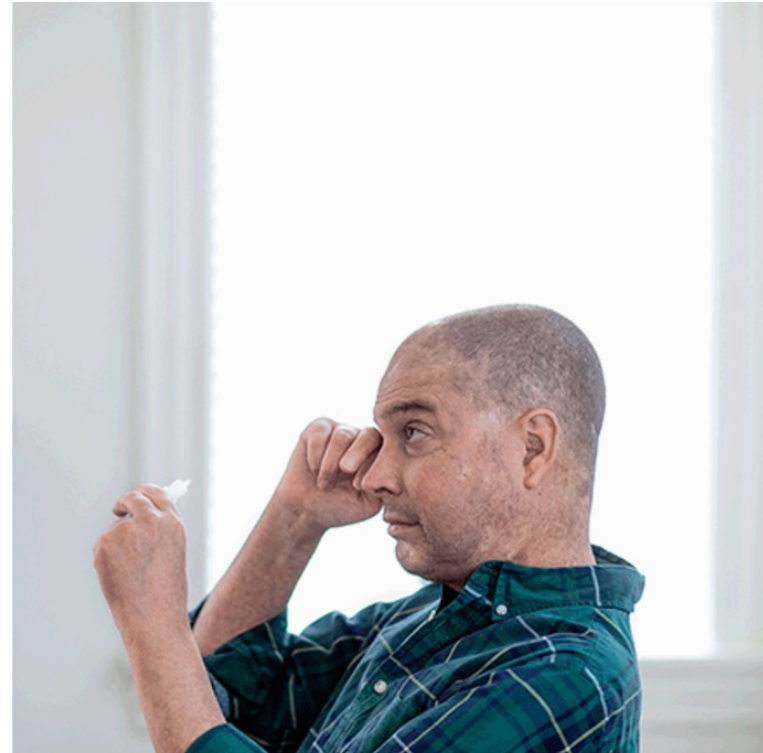


Treatment is still tough

A stem cell transplant helped beat back a young doctor's cancer. Now, it's assaulting his body

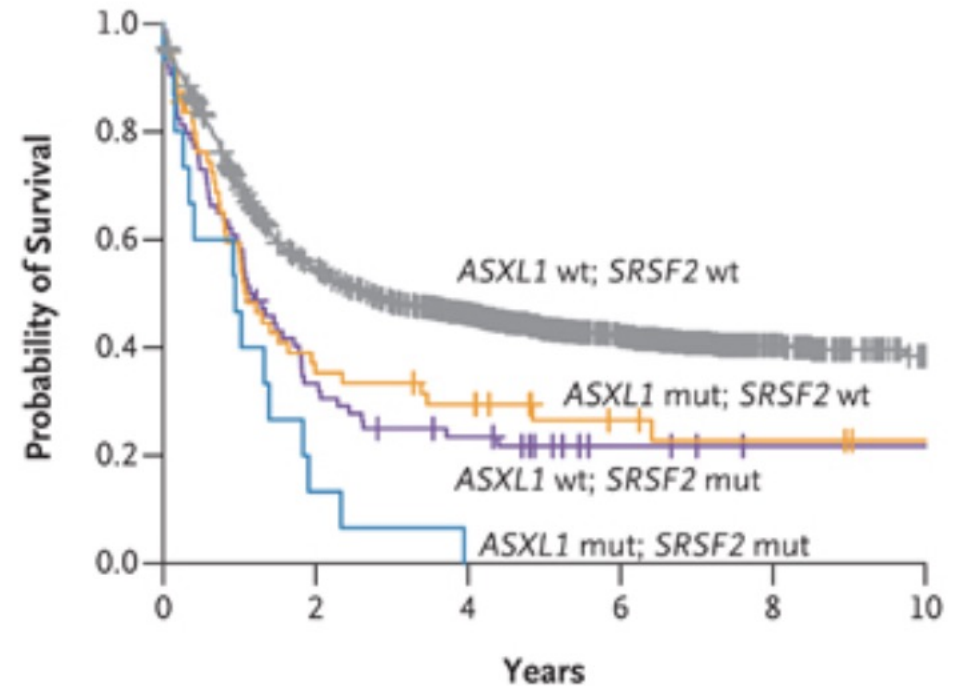
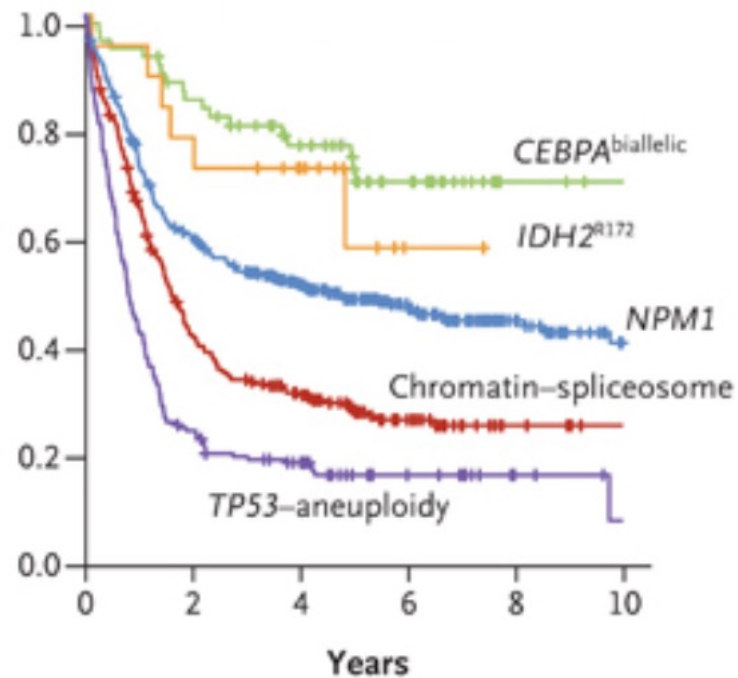
A few months before completing medical school in 2003, Lukas Wartman was diagnosed with acute lymphoblastic leukemia (ALL), a blood cancer that's particularly lethal when it strikes adults. So began a battle to stay alive that has involved more than 70 drugs, two rounds of cell transplants, and a staggering series of twists and turns.

Wartman immediately received aggressive chemotherapy, which drove the cancer into remission and enabled him to graduate from Washington



Still work to do

- Genomic medicine is here



- Too often, no treatment options

Still work to do

- Genomic medicine requires expertise in bioinformatics
- Building pipelines, crunching data
- Interpreting results
- Curating databases

The screenshot displays the CIViC website interface for the BRAF gene. The left sidebar contains navigation menus for Knowledgebase, Curation, Community, and Resources. The main content area shows the BRAF gene summary, including a description of its function as a B-Raf proto-oncogene, a list of sources (PubMed articles), aliases (B-RAF1, BRAF-1, etc.), and molecular profiles. The description states: "BRAF mutations are found to be recurrent in many cancer types. Of these, the mutation of valine 600 to glutamic acid (V600E) is the most prevalent. V600E has been determined to be an activating mutation, and cells that harbor it, along with other V600 mutations are sensitive to the BRAF inhibitor dabrafenib. It is also common to use MEK inhibition as a substitute for BRAF inhibitors, and th..." The sources section lists three PubMed articles: "Li et al., 2009", "Pakneshan et al., 2013", and "Yao et al., 2017". The aliases section lists B-RAF1, BRAF-1, B-raf, BRAF1, and NS7. The molecular profiles section shows a filter for "Molecular Profile Names" and a "Show" dropdown set to "All Molecular Profiles". Below this, there are buttons for various BRAF variants: BRAF V600E, BRAF V600, BRAF Mutation, BRAF KIAA1549::BRAF, BRAF V600K, BRAF V600D, BRAF Non-V600, BRAF D594G, and BRAF D594K.

CIViC
CLINICAL INTERPRETATION OF
VARIANTS IN CANCER

Search CIViC Home About CIViC Help Documentation Sign In / Sign Up

KNOWLEDGEBASE

- Assertions
- Evidence
- Molecular Profiles
- Features**
- Variants
- Variant Groups
- Clinical Trials
- Diseases
- Therapies
- Phenotypes
- Sources
- Variant Types

CURATION

- Activity
- Queues

COMMUNITY

- Contributors
- Organizations

RESOURCES

- Data Releases

Features / BRAF / Summary

GENE **BRAF** B-Raf proto-oncogene, serine/threonine kinase

Summary Comments Revisions (2) Flags Events Curators: Editors:

Description

BRAF mutations are found to be recurrent in many cancer types. Of these, the mutation of valine 600 to glutamic acid (V600E) is the most prevalent. V600E has been determined to be an activating mutation, and cells that harbor it, along with other V600 mutations are sensitive to the BRAF inhibitor dabrafenib. It is also common to use MEK inhibition as a substitute for BRAF inhibitors, and th... [Expand](#)

Sources

- PubMed: Li et al., 2009
- PubMed: Pakneshan et al., 2013
- PubMed: Yao et al., 2017

Aliases

- B-RAF1
- BRAF-1
- B-raf
- BRAF1
- NS7
- RAFB1

Resources

- DGIdb
- ProteinPaint

MyGeneInfo

[Overview](#) Summary Protein Domains (...)

Entrez Symbol: BRAF (ID: 673)

UniProtKB ID: P15056

Chromosome: 7 **Strand:** -1

Start: 140419127 **Stop:** 140624564

Aliases

B-RAF1, B-raf, BRAF1, NS7, RAFB1

Protein Domains

Diacylglycerol/phorbol-ester binding, Protein kinase C-like, phorbol ester/diacylglyc... [Expand](#)

Pathways

Downstream signaling in naïve CD8+ T cells, CDC42 signaling events, ErbB1 ... [Expand](#)

Molecular Profiles Variants

Filter: Molecular Profile Names Show: All Molecular Profiles

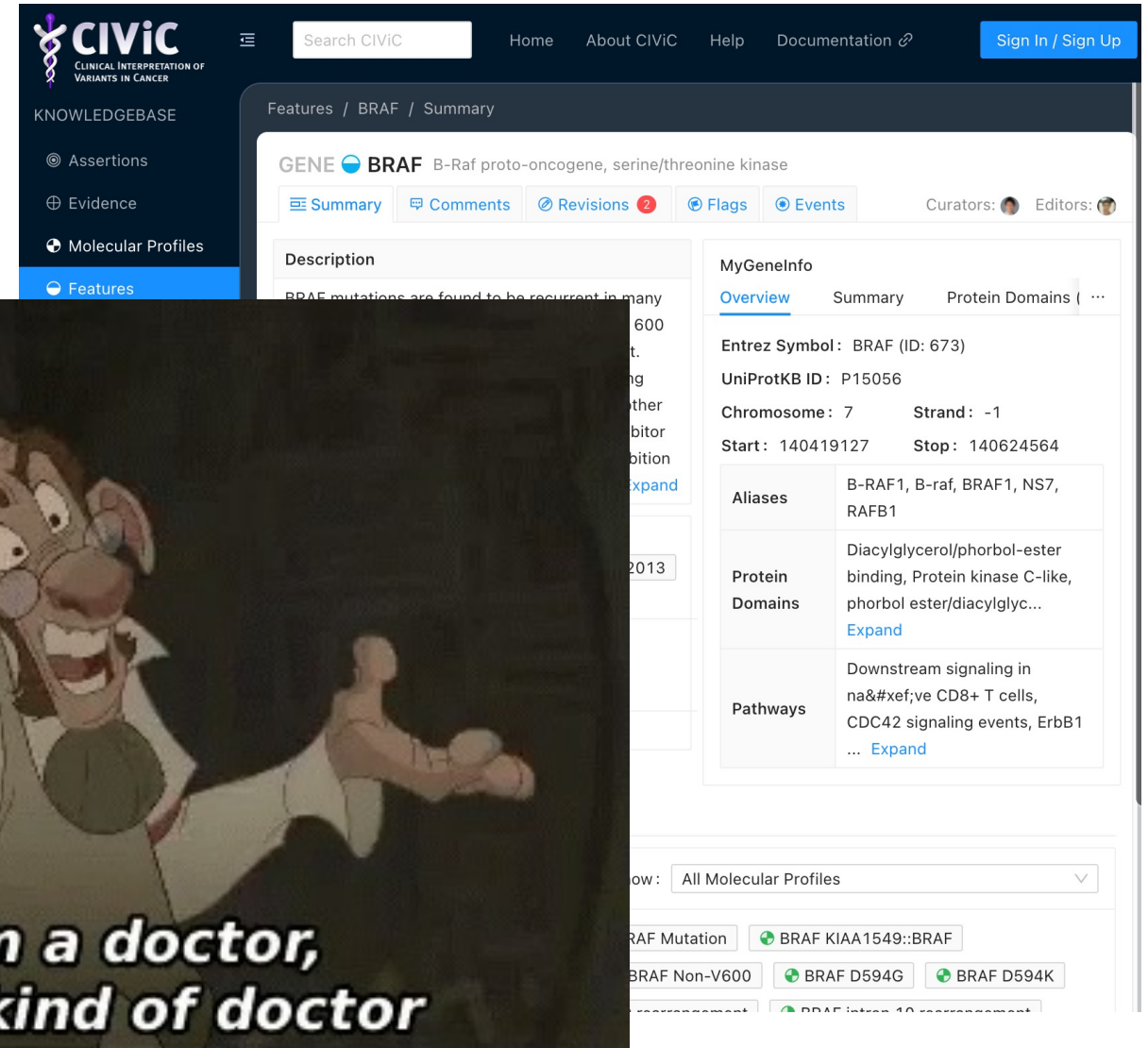
BRAF V600E BRAF V600 BRAF Mutation BRAF KIAA1549::BRAF

BRAF V600K BRAF V600D BRAF Non-V600 BRAF D594G BRAF D594K

BRAF A480::BRAF BRAF intron 2 splicing BRAF intron 10 splicing

Still work to do

- Genomic medicine requires expertise in bioinformatics
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CIViC
CLINICAL INTERPRETATION OF
VARIANTS IN CANCER

Search CIViC Home About CIViC Help Documentation Sign In / Sign Up

KNOWLEDGEBASE

- Assertions
- Evidence
- Molecular Profiles
- Features**

Features / BRAF / Summary

GENE **BRAF** B-Raf proto-oncogene, serine/threonine kinase

Summary Comments Revisions (2) Flags Events Curators: Editors:

Description

BRAF mutations are found to be recurrent in many...

MyGeneInfo

- Overview** Summary Protein Domains (...)
- Entrez Symbol:** BRAF (ID: 673)
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- Chromosome:** 7 **Strand:** -1
- Start:** 140419127 **Stop:** 140624564
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- Protein Domains:** Diacylglycerol/phorbol-ester binding, Protein kinase C-like, phorbol ester/diacylglyc... [Expand](#)
- Pathways:** Downstream signaling in naïve CD8+ T cells, CDC42 signaling events, ErbB1 ... [Expand](#)

How: All Molecular Profiles

RAF Mutation [BRAF KIAA1549::BRAF](#)

[BRAF Non-V600](#) [BRAF D594G](#) [BRAF D594K](#)

[BRAF inter 10](#)

**I mean I am a doctor,
but not that kind of doctor**

Course Review

- Bioinformatics is a journey, not a destination



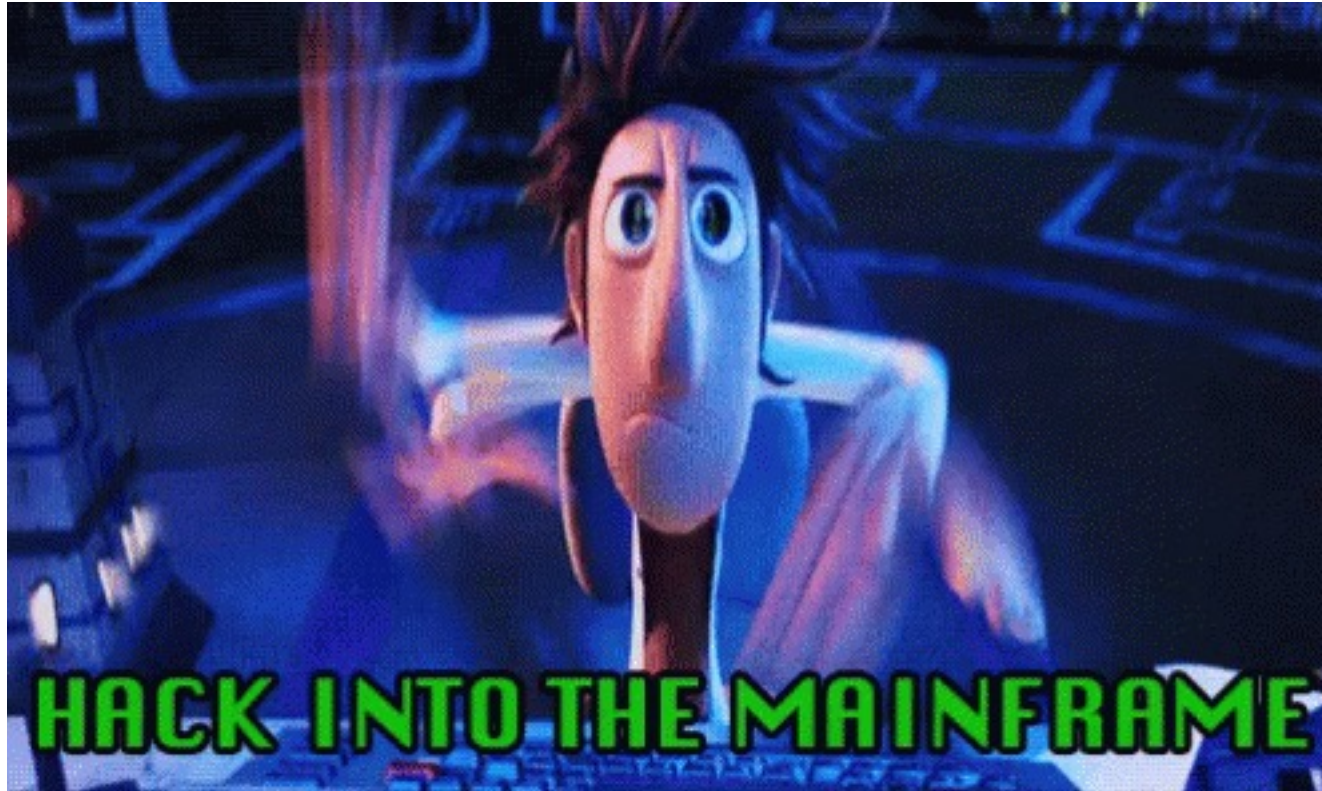
Course Review

- Shallow but broad, instead of deep and focused



Course Review

- The goal wasn't to turn everyone into computational experts



Course Review

- Give you familiarity with tools, conventions, and data types



Foundational skills

- Command line skills
- How to install packages in R or python
- How to find and use a docker container with the tools you need

```
acer@acer-PC MINGW64 ~/Desktop/New folder/Code
$ chmod +x shellscripts/demo.sh

acer@acer-PC MINGW64 ~/Desktop/New folder/Code
$ ./shellscripts/demo.sh
Hello from BASH
A very basic script
This is some text echoed

acer@acer-PC MINGW64 ~/Desktop/New folder/Code
$ ./demo.sh
bash: ./demo.sh: No such file or directory

acer@acer-PC MINGW64 ~/Desktop/New folder/Code
$
```



Foundational skills

- How to run things in different computing environments



Laptop



Cluster



Cloud

Foundational skills

- Understanding raw data formats and files
 - FASTQ, BAM, CRAM, BED, GTF, VCF



A place to start

- An idea of first steps to take with your data
- QC, Alignment, feeding into workflows



Foundational skills

- Exposure to different tools and techniques that you can apply
 - "This looks like a problem that we can answer with scRNA-seq!"

Foundational skills

- Exposure to different tools and techniques that you can apply
 - "This looks like a problem that we can answer with scRNA-seq!"
 - "This is **not** a problem that we can answer with scRNA-seq!"

Foundational skills

- Ideas of what happens when something goes wrong
 - Interpreting error messages
 - Troubleshooting experimental data
 - Using controls in your computational experiments

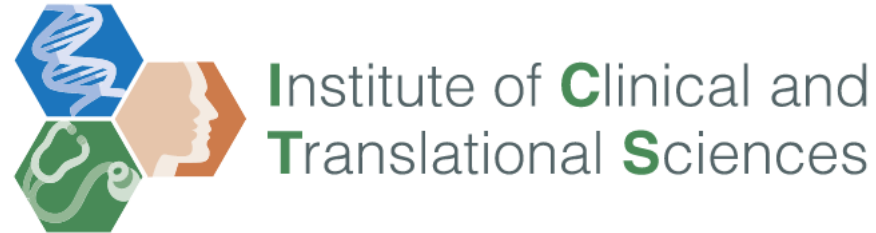
Foundational skills

- Ideas of what happens when something goes wrong
 - Interpreting error messages
 - Troubleshooting experimental data
 - Using controls in your computational workflows
- Don't trust your data until you have no other choice!

Thanks

- Jason Walker
- Susanna Kiwala
- Kartik Singhal
- My Hoang
- Mariam Khanfar
- Jennifer Foltz
- Juan Macias
- Brigida Rusconi
- Malachi Griffith

- Jenny McKenzie
- John Garza



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